Ensemble Quantum Computing Laboratory

Prelab 3

Runzhao Guo

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Problem 1

part a).

The spectra is shown in Figure 1.

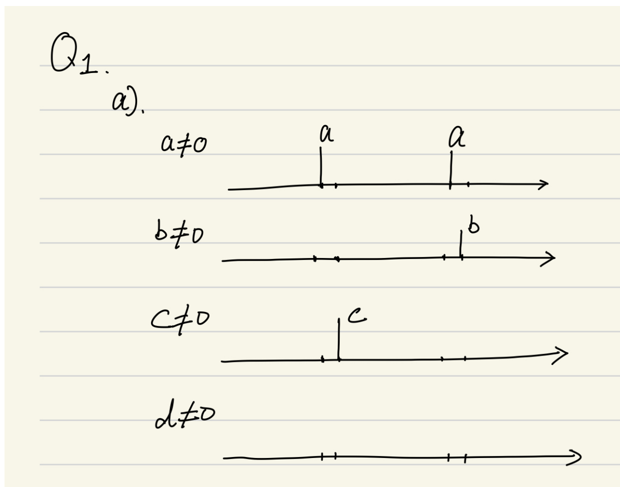

Figure 1: the spectra for the four cases when only one of a, b, c, and d is nonzero

part b).

The integral of the given thermal equilibrium is shown in Figure 2. Notice the frequence increase from left to right.

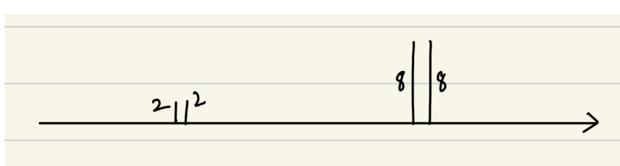

Figure 2: the integral for given thermal equilibrium density matrix

part c).

For CNOT gate, the density matrix is processed as such:$\rho = U_{cn}\rho_{\text{therm}}\, U_{cn}^{\dagger}$, the result we get is:

$$\begin{pmatrix} 1/4+5\times10^{-4} & 0 & 0 & 0 \\ 0 & 1/4+3\times10^{-4} & 0 & 0 \\ 0 & 0 & 1/4-5\times10^{-4} & 0 \\ 0 & 0 & 0 & 1/4-3\times10^{-4} \end{pmatrix}$$

For near-CNOT gate, the density matrix is processed as such:$\rho = U_{ncn}\rho_{\text{therm}}\, U_{ncn}^{\dagger}$, the result we get is:

$$\begin{pmatrix} 1/4-i5\times10^{-4} & 0 & 0 & 0 \\ 0 & 1/4+i3\times10^{-4} & 0 & 0 \\ 0 & 0 & 1/4+5\times10^{-4} & 0 \\ 0 & 0 & 0 & 1/4+3\times10^{-4} \end{pmatrix}$$

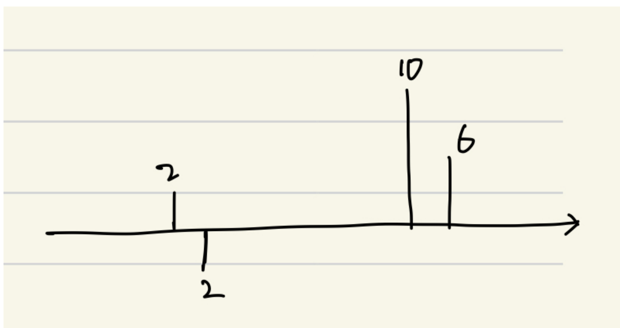

Figure 3: the integral for given thermal equilibrium density matrix after CNOT gate

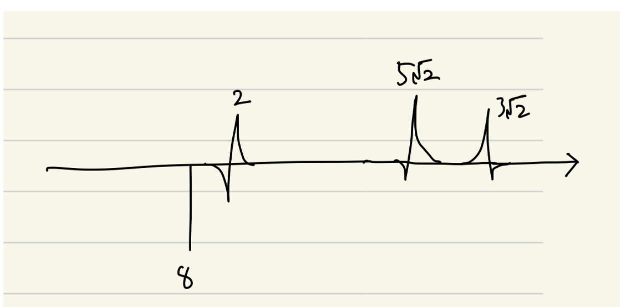

Figure 4: the integral for given thermal equilibrium density matrix after near-CNOT gate

The reletive relation between the integral of two peeks for Proton remain the same.

Problem 2

part a).

For a given Hamiltonian, the thermal equilibrium density matrix ρ_{eq} is given by:

$$\rho_{eq} = \frac{e^{-\beta H}}{Tr[e^{-\beta H}]}$$

where $\beta = \frac{1}{K_B T}$

The Hamiltonian for this two-qubit system is $\hat{H} = -\hbar\omega_H(1+\delta_H)\hat{I}_Z - \hbar\omega_C(1+\delta_C)\hat{S}_Z + 2\pi J\hat{I}_Z\hat{S}_Z$

Thank goodness it's a diagonal matrix, so the exponential can be calculated directly:

$$\rho_{eq} = \frac{1}{\sum_{i=1}^{4} e^{-\beta E_i}} \begin{pmatrix} e^{-\beta E_1} & 0 & 0 & 0 \\ 0 & e^{-\beta E_2} & 0 & 0 \\ 0 & 0 & e^{-\beta E_3} & 0 \\ 0 & 0 & 0 & e^{-\beta E_4} \end{pmatrix}$$

Where:
$E_1 = -0.5\hbar(\omega_H(1+\delta_H) + \omega_C(1+\delta_C) - \pi J)$
$E_2 = -0.5\hbar(\omega_H(1+\delta_H) - \omega_C(1+\delta_C) + \pi J)$
$E_3 = +0.5\hbar(\omega_H(1+\delta_H) - \omega_C(1+\delta_C) - \pi J)$
$E_4 = +0.5\hbar(\omega_H(1+\delta_H) + \omega_C(1+\delta_C) + \pi J)$
After calculation, the thermal equilibrium density matrix is:

$$\frac{1}{4.000 + 5.66 \times 10^{-11}} \begin{pmatrix} 1 + 6.306 \times 10^{-6} & 0 & 0 & 0 \\ 0 & 1 + 3.771 \times 10^{-6} & 0 & 0 \\ 0 & 0 & 1 - 3.771 \times 10^{-6} & 0 \\ 0 & 0 & 0 & 1 - 6.306 \times 10^{-6} \end{pmatrix}$$

part b

In my understanding, expressed in product operators, the density matrix is:

$$0.25\hat{I_I}\hat{S_I} + 0.5 \times 10^{-5}\hat{I_I}\hat{S_Z} + 0.13 \times 10^{-5}\hat{I_Z}\hat{S_I}$$

Expressing it in a similar way used in Knill paper:

$$\frac{1}{4.000} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + 10^{-5} \begin{pmatrix} 0.157 & 0 & 0 & 0 \\ 0 & 0.094 & 0 & 0 \\ 0 & 0 & -0.094 & 0 \\ 0 & 0 & 0 & -0.157 \end{pmatrix}$$

In comparison, the second element in my density matrix is 1/6 of that in the Knill paper. One possible reason is The magnetic field used in the Knill paper is stronger than that in my case, which leads to a larger value of $\omega_{H/C}$.

Problem 3

The exaustive average is:

$$\begin{pmatrix} 0.25 + 1.57 \times 10^{-6} & 0 & 0 & 0 \\ 0 & 0.25 - 3.2 \times 10^{-7} & 0 & 0 \\ 0 & 0 & 0.25 - 3.2 \times 10^{-7} & 0 \\ 0 & 0 & 0 & 0.25 - 3.2 \times 10^{-7} \end{pmatrix}$$

The excess probability is $\rho_{00} - \bar{p} = 1.89 \times 10^{-6}$

Problem 4

part a).

$$\mathrm{CNOT}_{(1\to 2)} = \frac{1}{2}\Big(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X\Big).$$

$$|0\rangle\langle 0| = \frac{1}{2}(I+Z)$$

$$|1\rangle\langle 1| = \frac{1}{2}(I-Z)$$

$$\mathrm{CNOT}_{(1\to 2)} = \frac{1}{2}\Big(I\otimes I \;+\; Z\otimes I \;+\; I\otimes X \;-\; Z\otimes X\Big).$$

$$= \frac{1}{4}\begin{pmatrix} 1+1 & 1-1 & 0 & 0 \\ 1-1 & 1+1 & 0 & 0 \\ 0 & 0 & 1-1 & 1+1 \\ 0 & 0 & 1+1 & 1-1 \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

part b).